OpenMP Target Offload for Heterogeneous Architectures

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Goal: Unified *open high-level* programming of both CPU & GPU
Easy yet high-performance



Exaflop/s Supercomputing

- **Diverse Exaflop/s supercomputing platforms**



Summit (0.2 Exaflop/s)
IBM CPU/NVIDIA GPU



1 HPC and AI Optimized
AMD EPYC CPU
4 Purpose Built AMD
Radeon Instinct GPU



Frontier (1.7 Exaflop/s)
AMD CPU/AMD GPU



GPU Architecture

X^e arch-based "Ponte Vecchio"
GPUTile-based, chiplets, HBM
stack, Foveros 3D integration, 7nm

On-Node Interconnect

CPU-GPU: PCIe
GPU-GPU: Xe Link

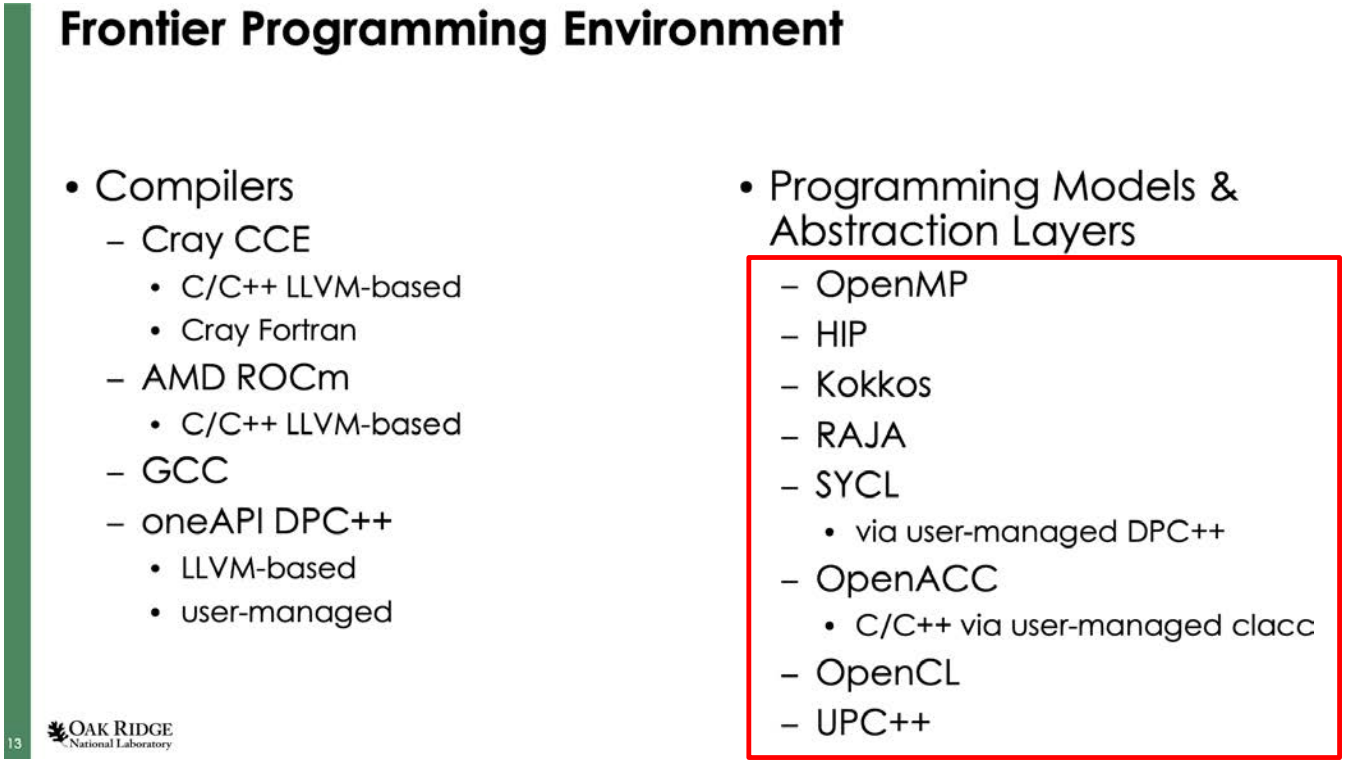


Aurora (2.0 Exaflop/s)
Intel CPU/Intel GPU

- **Need an *open* programming model for *heterogeneous* (*e.g.*, GPU-accelerated) clusters (note CUDA is a proprietary language by NVIDIA)**

See https://atpesc-archive.anl.gov/agenda-2024/

Frontier (AMD) Programming



No CUDA

See https://atpesc-archive.anl.gov/agenda-2024/

Aurora (Intel) Programming

Aurora Programming Models

- Aurora applications may use
  - DPC++/SYCL
  - OpenMP
  - Kokkos
  - Raja
  - OpenCL
- Experimental
  - HIP
- Not available on Aurora
  - CUDA
  - OpenACC

No CUDA



See https://atpesc-archive.anl.gov/agenda-2024/

Open Programming Models

- **OpenACC (Open Computing Language)**

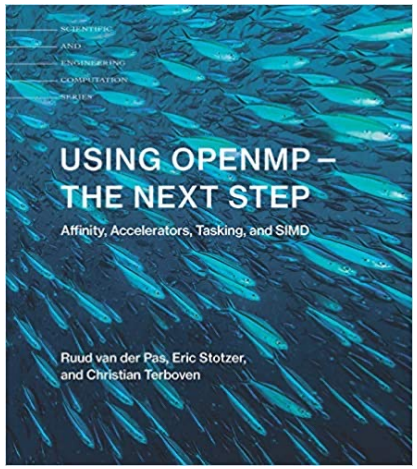
  Open standard for directive-based programming of heterogeneous devices

  https://www.openacc.org/

- **OpenMP 4.5/5**

  Starting specification version 4.5, OpenMP allows offloading the execution of code & data to heterogeneous devices

  https://www.openmp.org/specifications/



https://www.amazon.com/Using-OpenMP_The-Next-Step-Accelerators/dp/0262534789

OpenMP Offload

- **Latest version of OpenMP allows one to maintain one version of a code, which can run on either a general-purpose central processing unit (CPU) or an accelerator (*e.g.* graphics processing unit, GPU; tensor processing unit, TPU; digital signal processor, DSP; field-programmable gate array, FPGA)**
- **Objective is to execute parts of the program on a heterogeneous *accelerator device* (or *target device*), *i.e.*, dedicated computer hardware outside CPU (which will be called *host device*) to execute certain functions faster than CPU**
- **In OpenMP, program execution begins on the host, which offloads the execution of parts of the code & data to accelerator**

See “OpenMP 4.5 Target Offload” (NASA Ames)
https://aiichironakano.github.io/cs653/OpenMP4.5_3-20-19.pdf

OpenMP Target Construct

- **Simple example**

```
main() {
  float a[1000],b[1000],c,d;
  ...
  #pragma omp target map(a,b,c,d)
  {
    #pragma omp parallel for
    for (int i=0; i<N; i++)
      a[i] = b[i]*c+d;
  }
  ...
}
```

One-line GPU offload!

Same on CPU or GPU

- **When a host thread encounter the `#pragma omp target` directive, the target region specified by it will be executed by a new thread running on an accelerator, *cf.* CUDA GPU kernel**
- **Before the new thread starts executing the target region, the variable in the `map()` clause are mapped onto accelerator memory, which often is disjunct from the host memory, *cf.* `cudaMemcpy()`**
- **The offloaded code is usually a data-parallel structured block, which can be handled by multiple threads on accelerator using standard OpenMP constructs like `#pragma omp parallel for`**

Computing the Value of π on GPU

omp_target_pi.c

```
#include <omp.h>
#include <stdio.h>
#define NBIN 1000000
#define NTRD 96

int main() {
  float step,sum=0.0,pi;
  step = 1.0/(float)NBIN;

  #pragma omp target map(step,sum)
  {
    # pragma omp parallel for reduction(+:sum) num_threads(NTRD)
    for (long long i=0; i<NBIN; i++) {
      float x = (i+0.5)*step;
      sum += 4.0/(1.0+x*x);
    }
  }

  pi = sum*step;
  printf("PI = %f\n",pi);
  return 0;
}
```

The only addition for GPU offload

Thread reduction of sum

Specify # of GPU threads

This line is identical for CPU & GPU

1. **Black: original serial code**
2. **Green: one-line multithreading**
3. **Red: another line for GPU offload**

GPU: Easy & Hard Ways

Serial

```
#include <stdio.h>
#define NBIN 1000000000
int main() {
   double step, x, sum=0.0, pi;
   step = 1.0/NBIN;
   for (long long i=0; i<NBIN; i++) {
      x = (i+0.5)*step;
      sum += 4.0/(1.0+x*x);
   }
   pi = sum*step;
   printf("PI = %f\n",pi);
   return 0;
}
```

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OpenMP

```
#include <omp.h>
#include <stdio.h>
#define NBIN 1000000
int main() {
   float step,sum=0.0,pi;
   step = 1.0/(float)NBIN;
   #pragma omp target map(step,sum)
   {
      # pragma omp parallel for reduction(+:sum)
      for (long long i=0; i<NBIN; i++) {
         float x = (i+0.5)*step;
         sum += 4.0/(1.0+x*x);
      }
   }
   pi = sum*step;
   printf("PI = %f\n",pi);
   return 0;
}
```

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SYCL: pi.cpp

```
#include <CL/sycl.hpp>
#include <iostream>
#include <array>

using namespace cl::sycl;

#define NBIN 1000000  // # of bins for quadrature
#define NTRD 512      // # of threads

int main()
{
  float step = 1.0f/NBIN;
  std::array<float, NTRD> sum;
  for (int i=0; i<NTRD; ++i) sum[i] = 0.0f;

  queue q(gpu_selector{});

  std::cout << "Running on: " <<
  q.get_device().get_info<info::device::name>() << std::endl;

  range<1> sizeBuf{NTRD};

  {
    buffer<float, 1> sumBuf(sum.data(), sizeBuf);
    q.submit([&](handler &h){
      auto sumAccessor =
      sumBuf.get_access<access::mode::read_write>(h);
      h.parallel_for(sizeBuf, [=](id<1> tid) {
        for (int i=tid; i<NBIN; i+=NTRD) {
          float x = (i+0.5f)*step;
          sumAccessor[tid] += 4.0f/(1.0f+x*x);
        }
      });  // End parallel for
    });  // End queue submit
  }

  float pi=0.0f;
  for (int i=0; i<NTRD; i++)  // Inter-thread reduction
    pi += sum[i];
  pi *= step;  // Multiply bin width to complete integration

  std::cout << "Pi = " << pi << std::endl;

  return 0;
}
```

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CUDA: pi.cu

```
// Using CUDA device to calculate pi
#include <stdio.h>
#include <cuda.h>

#define NBIN  10000000  // Number of bins
#define NUM_BLOCK   13  // Number of thread blocks
#define NUM_THREAD 192  // Number of threads per block
int tid;
float pi = 0;

// Kernel that executes on the CUDA device
__global__ void cal_pi(float *sum, int nbin, float step, int nthreads, int nblocks) {
   int i;
   float x;
   int idx = blockIdx.x*blockDim.x+threadIdx.x;  // Sequential thread index across the blocks
   for (i=idx; i< nbin; i+=nthreads*nblocks) {
      x = (i+0.5)*step;
      sum[idx] += 4.0/(1.0+x*x);
   }
}

// Main routine that executes on the host
int main(void) {
   dim3 dimGrid(NUM_BLOCK,1,1);  // Grid dimensions
   dim3 dimBlock(NUM_THREAD,1,1);  // Block dimensions
   float *sumHost, *sumDev;  // Pointer to host & device arrays

   float step = 1.0/NBIN;  // Step size
   size_t size = NUM_BLOCK*NUM_THREAD*sizeof(float);  //Array memory size
   sumHost = (float *)malloc(size);  //  Allocate array on host
   cudaMalloc((void **) &sumDev, size);  // Allocate array on device
   // Initialize array in device to 0
   cudaMemset(sumDev, 0, size);
   // Do calculation on device
   cal_pi <<<dimGrid, dimBlock>>> (sumDev, NBIN, step, NUM_THREAD, NUM_BLOCK); // call CUDA kernel
   // Retrieve result from device and store it in host array
   cudaMemcpy(sumHost, sumDev, size, cudaMemcpyDeviceToHost);
   for(tid=0; tid<NUM_THREAD*NUM_BLOCK; tid++)
      pi += sumHost[tid];
   pi *= step;

   // Print results
   printf("PI = %f\n",pi);

   // Cleanup
   free(sumHost);
   cudaFree(sumDev);

   return 0;
}
```

Hierarchical Parallelization

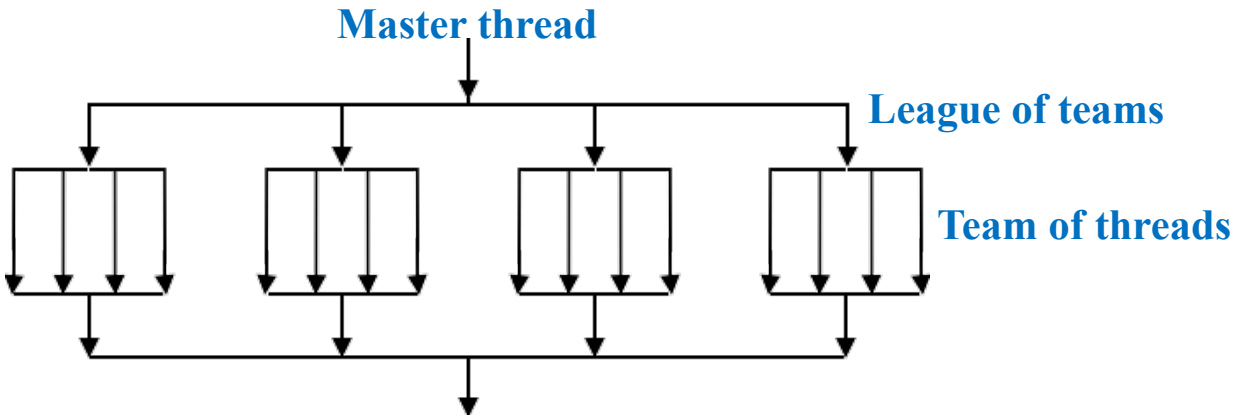
- `#pragma omp teams`

  Starts a league of multiple thread teams; `teams` **construct must be nested immediately inside a** `target` **construct, so they are commonly used as** `#pragma omp target teams`**;** `num_teams` **clause can be used to specify the number of teams**

  (Example) `#pragma omp target teams num_teams(13)`

- `#pragma omp distribute`

  Distribute the work across the teams



cf. **CUDA grid of blocks & block of threads**

Remember NVIDIA SM/SP & Intel slice/stack?

Teams for Computing π

- **Spatial decomposition *via* offset among teams & data privatization**

```
#define NTMS 12
float sum_teams[NTMS];
for (int j=0; j<NTMS; j++) sum_teams[j] = 0.0;
...
#pragma omp target teams map(step,sum_teams) num_teams(NTMS)
{
  #pragma omp distribute
  for (int j=0; j<NTMS; j++) {
    long long ibgn = NBIN/NTMS*j;
    long long iend = NBIN/NTMS*(j+1);
    if (j == NTMS-1) iend = NBIN;
    # pragma omp parallel for reduction(+:sum_teams[j]) num_threads(NTRD)
    for (long long i=ibgn; i<iend; i++) {
      float x = (i+0.5)*step;
      sum_teams[j] += 4.0/(1.0+x*x);
    }
  }
}

for (int j=0; j<NTMS; j++) sum += sum_teams[j];
```

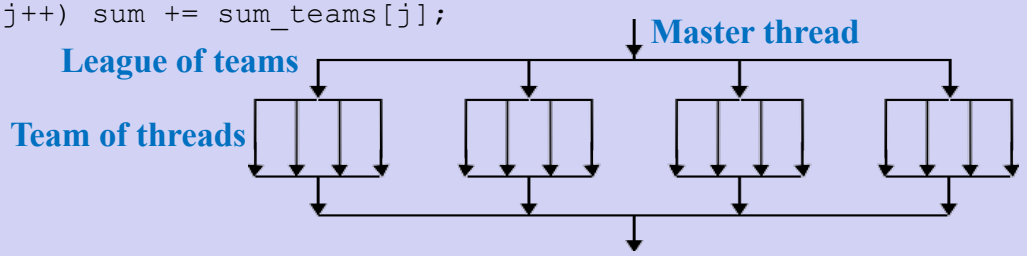
Data privatization among teams

New: teams & distribute constructs

⌊NBIN/NTMS⌋×NTMS could be less than NBIN

Modified: offset & private accumulator

Interteam reduction



Using OpenMP Target on Discovery

- **Necessary module**

```
module purge
module load nvhpc
```

- **Compilation**

```
nvc -o omp_target_pi omp_target_pi.c
```

NVIDIA C compiler supports newer C constructs

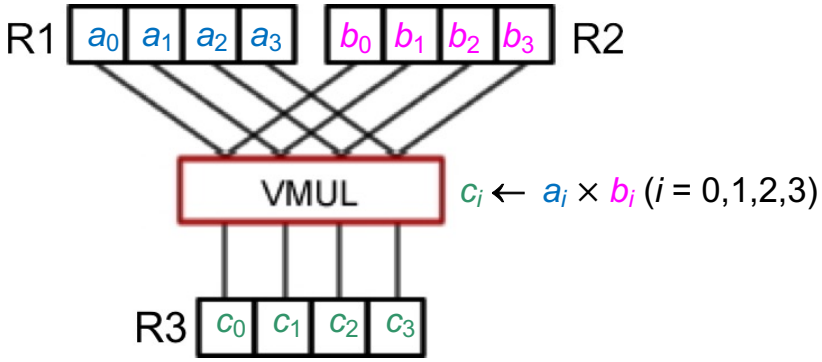
- **Execute on a GPU-accelerated node**

```
[anakano@discovery]$ salloc --partition=gpu --time=00:30:00
salloc: Nodes e22-05 are ready for job
[anakano@e22-05]$ ./omp_target_pi
PI = 3.141593
```

Single Instruction Multiple Data

- **OpenMP 4.5 & later supports several other new features**
- **Single-instruction multiple-data (SIMD) parallelism: An arithmetic operation is operated on multiple operand-pairs stored in vector registers (each of which can hold multiple operands) using vector instructions**
- **OpenMP `simd` construct instructs the compiler to vectorize the loop**

```
#pragma omp for simd
for (int i=0; i<n; i++)
  a[i] = b[i]+c[i];
```



Asynchronous Offload

```
main() {
  float a[1000],b[1000],c,d;
  ...
  #pragma omp target nowait map(a,b,c,d)
  {
    #pragma omp parallel for
    for (int i=0; i<N; i++)
      a[i] = b[i]*c+d;
  }
  func(b);   // perform computation independent of device output
  #pragma omp taskwait
  func(a);   // perform computation dependent on device output
}
```

cf. MPI_Irecv() & MPI_Wait()

- **By default, the thread that encounters a device construct waits for the construct to complete before executing the next line**
- **When a `nowait` clause is added to the device construct, the encountering thread does not wait but instead continues executing the code passed the construct**
- **The `taskwait` constructs lets the original thread wait for the completion of the target task generated by it before continuing to the next line**

Overlap CPU & GPU computations for high performance

See Porting LFD Mini-app to GPU via OpneMP Offload (Pankaj Rajak, USC/ANL)

Persistent GPU Memory Allocation

- **Expensive CPU-GPU data transfer associated with the `map` clause can be minimized by making GPU memory allocation persistent**

cf. **3-phase (host-to-device copy→kernel execution→device-to-host copy) performance bottleneck**

```
// Pre-allocate GPU array
#pragma omp target enter data map(alloc:psi[0:Nmax])

...

// Keep operating on device array only from device
#pragma omp target parallel for \
map(tofrom:psi[0:Nmax]) map(to:u[0:Nmax])
for (int i=0; i< Nmax; i++) psi[i] *= u[i];

...

// De-allocate GPU array
#pragma omp target exit data map(delete:psi[0:Nmax])
```

Stand-alone directive to map variables to device memory

to: host-to-device
from: device-to-host

Runtime system keeps track of CPU & GPU memory access & avoids unnecessary CPU-GPU data transfer

Stand-alone directive to unmap variables from device memory

Used in DCMSH code T. M. Razakh *et al.*, *SC25*, doi: 10.1145/3712285.3771785

Putting All Together to Achieve Multi-Exaflop/s

- **Open standard: Portable across supercomputers through MPI (message passing) + OpenMP (multithreading) + OpenMP target (GPU)**
- **Data/loop reordering, blocking & hierarchical parallelization: Reuse spatial stencil operators for an array of electronic wave functions as structure of arrays (SoA), along with blocking for cache utilization & hierarchical parallelization over teams & threads**
- **GPU resident kernels: Shadow dynamics (MSA) makes large electronic wave functions GPU-resident to minimize CPU-GPU data transfer, facilitated by customized OpenMP target data allocator/destructor to ease programming**
- **Parametrized mixed-precision computation through class template with parameterized precision for FP64 QXMD, FP32 LFD to compute small occupation number ($\in [0, 1]$) correction, and FP32 GEMM emulation using BF16 systolic arrays when computing perturbatively small nonlocal correction in LFD**

cf. 48-bit gravitational pipe: Ito *et al*., *Comput. Phys. Commun*. **60**, 187 ('90)

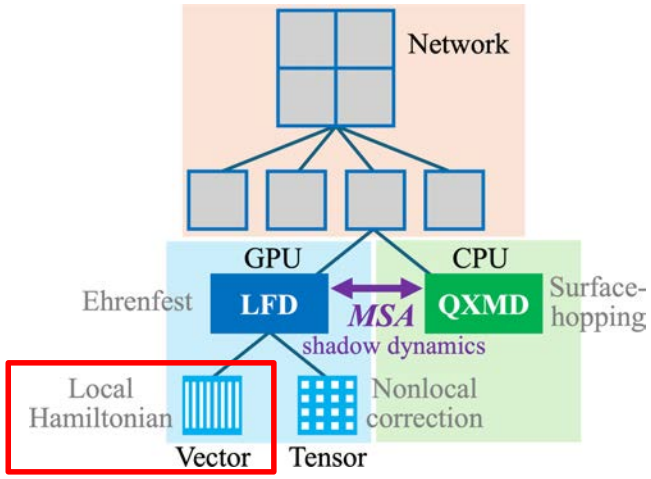
Achieved 1.87 Exaflop/s on 60,000 GPUs of Aurora

Multiscale light-matter dynamics in quantum materials,
T. M. Razakh *et al*., *2025 Gordon Bell prize finalist*



Local Time-Propagator Optimization

- **Data/loop reordering, blocking & hierarchical parallelization**
  1. **Reuse spatial stencil operators for an array of electronic wave functions each on N_{grid} spatial grid points as structure of arrays (SoA)**
  2. **Blocking for cache utilization**
  3. **Hierarchical parallelization over teams & threads on GPU**



Runtime of the kin_prop() function for local time-propagation in the LFD subprogram of the DC-MESH module

Implementation	Target	Runtime (s)	Speedup
Baseline	CPU*	8.655	1
Data & loop re-ordering	CPU	2.356	3.67
Blocking/tiling	CPU	0.939	9.22
Hierarchical parallel regions	GPU#	0.026	338

*2.8 GHz AMD EPYC Milan 7543 CPU single core & #Nvidia A100 GPU on Polaris at Argonne ALCF

GEMMification of Nonlocal Computation

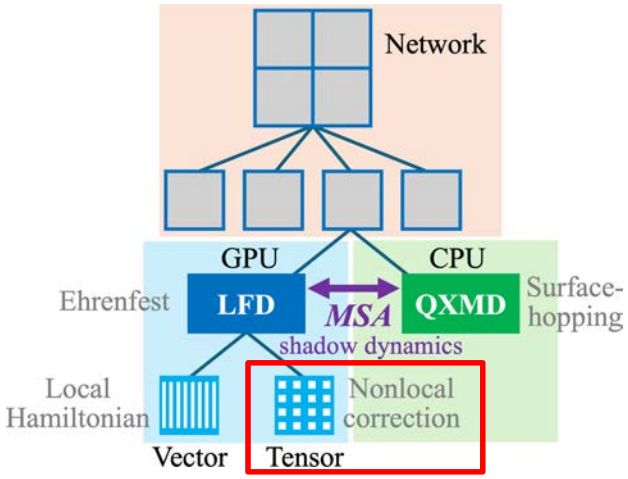
- **Switch of representation from finite-difference to Kohn-Sham (KS) orbital bases makes nonlocal correction into dense complex-matrix multiplications**

$$\overbrace{\Psi(t)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}} \mathrel{-}= \delta \overbrace{\Psi(0)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}} \overbrace{\Psi^{\dagger}(0)}^{N_{\mathrm{orb}}\times N_{\mathrm{grid}}} \overbrace{\Psi(t)}^{N_{\mathrm{grid}}\times N_{\mathrm{orb}}}$$

FP32 BF16

- **FP32 GEMM emulation with BF16**

N. Piroozan *et al.*, *SCW24*,
doi: 10.1109/SCW63240.2024.00187



Single GPU-tile performance of DC-MESH for several problem sizes*

Number of KS orbitals	TFLOP/s
256	5.22 (FP32)
864	9.74 (FP32)
1024	14.98 (FP32)
1024	**17.95 (FP32/BF16)**
1024	7.69 (FP64)

*Intel Data Center GPU Max 1550 GPU (PVC)

Single GPU-tile performance of hot-spot kernels

Kernels	TFLOP/s
CGEMM (1)	18.72 (FP32)
CGEMM (2)	21.66 (FP32)
nlp_prop()	16.02 (FP32)
kin_prop()	3.51 (FP32)

Local Nonlocal (GEMMified)

Where to Go from Here

- **Start developing your own OpenMP target offload codes for GPU acceleration (the easiest path)**
- **Plenty of room for performance optimization**

  ***e.g.*, target region executed by different threads happens concurrently, used for, *e.g.*, overlapping computation & data transfer**

```
#pragma omp parallel  // Spawn multiple CPU threads
{
  #pragma omp target
  {// Different GPU threads perform computation or data transfer }
}
```

See OpenMP Offload Optimization (Ye Luo, Argonne National Lab)

See also Accelerating quantum light-matter dynamics on graphics processing units,
T. M. Razach *et al.*, *Proc. IEEE IPDPSW/PDSEC* ('24)

Multiscale light-matter dynamics in quantum materials,
T. M. Razakh *et al.*, *2025 Gordon Bell prize finalist*